

Subhasri Vijay

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EDUCATION

JOHNS HOPKINS UNIVERSITY, Baltimore, Maryland

Master of Science in Engineering: Computer Science

Graduated May 2026

Bachelor of Science: Computer Science; Minors: Computational Medicine, Entrepreneurship and Management

Graduated May 2025

Relevant Coursework: Data Structures, Computer System Fundamentals, Algorithms, Deep Learning, Artificial Intelligence, Computational Genomics: Sequencing, Databases, Human Computer Interaction, Computer Vision, Machine Translation

SPECIALIZED SKILLS

- **Machine Learning & AI:** PyTorch, TensorFlow, Deep Learning, Computer Vision, Foundation Models, Model Training & Evaluation
- **Programming Languages:** Python, Java, C/C++, JavaScript, TypeScript, SQL
- **Web Development:** React, TypeScript, Full-Stack Development, HTML/CSS
- **Data & Analysis:** NumPy, Pandas, Data Preprocessing, Dataset Curation, Jupyter
- **Tools & Platforms:** Git, Linux, MATLAB, R

PROJECTS AND EXPERIENCE

Software Engineering Intern, Inflection Medicine

July 2026-Present

- Engineered gene/variant-level clinical-findings support in the Test Configuration Service (schema, REST API, validation, CLI), enabling seeded orders to generate reports combining seeded findings with order patient details; designed it route-agnostic to generalize across lab-fulfillment backends and to underpin product demos.
- Automated the first end-to-end test harness for the Clinical Guidance Engine (CGE): "Virtual Clinical Team", a headless agent driving the Curation and Guidance portal via APIs on TCS-seeded orders, designed to trigger the CGE workflow automatically whenever a seeded order appears and to run as a continuous mock for development and demos.
- Delivered the first curator-facing view over the CGE/PPS audit database: a reusable audit-history component spanning 16 record types across 8 entity pages, a global searchable Audit Log surfacing who changed what, and when.

Cataract Research Assistant, Wilmer Eye Institute

Jan 2022-May 2026

- Created and analyzed six surgeon-specific ground-truth strategies to benchmark model performance on the question: "Can AI accurately identify the expert surgeon?"; evaluated using accuracy, sensitivity, specificity, and precision. Findings presented at AUPO 2024.
- Curated and preprocessed 30+ cataract surgery videos in collaboration with clinicians to build high-quality, structured datasets for training and evaluating deep learning models.
- Applied and evaluated deep learning architectures (encoders, decoders) and metrics (AUC, confusion matrix) for surgical skill assessment, contributing to clinically relevant advancements in AI-driven surgical evaluation.

Co-Founder, RescueReady

Jan 2025-Sept 2025

- Co-founded and led development of an AI-powered EMT training platform that simulates high-pressure emergency scenarios to enhance decision-making and situational awareness.
- Built a full-stack web application using React, TypeScript, and SQLite, integrating generative AI to generate realistic, protocol-aligned simulations in collaboration with EMTs and healthcare professionals.
- Designed RESTful API architecture and implemented user authentication system to support multi-user training sessions with progress tracking.
- Achieved 3rd Place at HopStart General Ventures and won the Pava Center for Entrepreneurship Award, responsible for product vision, technical architecture, and user testing.

IMAZER (Image Processing Tool)

July 2024

- Developed a lightweight Python desktop application for batch image resizing with a Tkinter GUI, eliminating command-line dependency for non-technical users.
- Engineered support for multiple image formats with robust error handling for invalid files; optimized batch processing pipeline for reliable performance across large image sets.

POSTERS AND PUBLICATIONS

- Poster titled "Can AI accurately answer: Who is the expert surgeon?" at AUPO 2024.
- Poster titled "Surgical Skill Assessment of Cataract Videos" at ISBI 2025.
- Paper titled "A Vision Foundation Model for Cataract Surgery Using Joint-Embedding Predictive Architecture" at MIDL 2025.

TEACHING EXPERIENCE

Head Teaching Assistant, Data Structures: EN.601.226.

Feb 2024-May 2026

- Guided students in debugging complex programming tasks and simplified Data Structures concepts into accessible explanations, fostering better understanding, problem-solving, and technical communication skills.
- Conducted weekly office hours, providing hands-on assistance and promoting student engagement through practical learning experiences.
- Graded over 100 project submissions and 200 exam submissions while conducting 50+ office hours, ensuring fair grading and collaborating with faculty to create an inclusive and effective learning environment.

HONORS AND AWARDS

- Awarded Joel Dean Award for Excellence in Teaching by Computer Science department. 2026
- Dean's List for 8 semesters at Johns Hopkins University and graduated with General and Departmental Honors. 2025
- Awarded the Masson Fellowship for 2024 by the Computer Science department for Research. 2025